Om Anand

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EDUCATION

Kalinga Institute of Industrial Technology — Bhubaneswar, Odisha
B.Tech in Computer Science — CGPA: ***8.69 / 10.0*** — *Present*

JBM Public School (CBSE) — New Delhi
Higher Secondary Education — 76% — *2020 – 2022*

Jindal Public School (CBSE) — New Delhi
Secondary Education — 91% — *2010 – 2020*

RELEVANT COURSEWORK

- Data Structures & Algorithms
- Object-Oriented Programming
- Mobile Application Development
- Database Management Systems
- Operating Systems
- Automata & Formal Languages

TECHNICAL SKILLS

Programming Languages: Kotlin, Java, Python, C/C++, SQL, Bash
Mobile & Backend Dev: Jetpack Compose, Android SDK, Ktor 3.x, Koin (DI), JetBrains Exposed ORM, PostgreSQL, H2, WebSockets, JWT, OAuth 2.0, BCrypt, RESTful APIs
Web & Game Development: HTML5 Canvas, Vanilla JavaScript (ES Modules), Web Audio API, CSS3, Streamlit
Data & ML: NumPy, Pandas, Matplotlib, Scikit-learn, MLP Neural Networks, Power BI, Excel, Jupyter Notebook, NLP, TF-IDF
Networking & Systems: 5G Network Simulation, DASH Protocol, QoE Metrics, TCP/IP
Tools & Platforms: Android Studio, IntelliJ IDEA, VS Code, Git/GitHub, Gradle (Kotlin DSL), Google Workspace

RESEARCH WORK

ML-Based DASH Adaptation in 5G Networks — Ongoing
Research Project — Repositories: NASP, FiveGMetricsCollector

- Investigating ML-driven bitrate adaptation algorithms for DASH in 5G environments, focused on improving video QoE under bandwidth fluctuations, handovers, and mobility scenarios.
- Developed **FiveGMetricsCollector** — a custom pipeline to capture live 5G network metrics for training adaptive bitrate models; collaborating on **NASP** to benchmark ML-based vs. heuristic streaming strategies.

TECHNICAL PROJECTS

Active Flood Risk Management System — Completed
Technologies: Python, Scikit-learn, Streamlit, Pandas, NumPy, Matplotlib, Jupyter Notebook

- Processed **1.1M+ records**, engineering 20 variables into **7 Master Indices** to reduce dimensionality; benchmarked Linear Regression, Decision Tree, and **MLP Neural Network** via R^2, MSE, RMSE, MAE.
- Deployed an interactive **Streamlit dashboard** for real-time scenario simulation with an automated alert system classifying Low, Moderate, and High flood risk with actionable recommendations.

Graviton — Browser-Based Arcade Game — Completed
Technologies: Vanilla JavaScript (ES Modules), HTML5 Canvas, Web Audio API, CSS3

- Built a zero-dependency arcade runner with a procedurally generated spiral tunnel, irregular polygon asteroids, and a fixed-timestep **60 FPS physics engine**.
- Implemented an **AI Director** modulating difficulty via a dynamic stress level (0.0–1.0); engineered sine-wave drifting, fuel/health economy, time-stop boost, and procedural **Web Audio API** sound effects.

ThunderChat — Production-Grade Real-Time Chat Server — Completed
Technologies: Kotlin, Ktor 3.x, Koin, PostgreSQL, JetBrains Exposed ORM, JWT, Google OAuth 2.0, H2

- Architected a **Clean Architecture** Ktor backend with decoupled route/service/repository layers and **Koin** DI; implemented **JWT + Google OAuth 2.0** security and bidirectional WebSocket messaging.
- Built a full integration test suite with in-memory **H2** and live **PostgreSQL** persistence via JetBrains Exposed ORM.

MyPokedex — Secure Multi-Tenant REST API — eUdyaan — Mental Wellness App — Completed
Kotlin/Ktor, BCrypt, H2 — Kotlin, Jetpack Compose, Android Studio — NLP, Scikit-learn, TF-IDF (Sentiment Analyzer)

- **MyPokedex:** Multi-tenant REST API with strict data isolation, BCrypt session auth, H2 persistence, and parameterized PATCH/DELETE endpoints. **Sentiment Analyzer:** NLP pipeline with TF-IDF vectorization; evaluated via confusion matrices, ROC curves.
- **eUdyaan:** Mental wellness Android app with mood tracking, AI chatbot, anonymous forum, and consultation booking; prioritized **privacy-by-design** with **Jetpack Compose**.